

Chapter 3: Simplification of Boolean Functions

- The complexity of the digital logic gates that implement a Boolean function is directly related to the complexity of the algebraic expression from which the function is implemented.
- The procedure of minimization of Boolean functions by algebraic means is awkward because it lacks specific rules to predict each succeeding step in the manipulative process.
- The map method provides a simple straightforward procedure for minimizing Boolean functions.

References

1. Veitch, E. W., "A Chart Method for Simplifying Truth Functions," Proc. of the ACM (May 1952), 127-33.
2. Karnaugh, M., "A Map Method for Synthesis of Combinational Logic Circuits," Trans. AIEE, Comm. and Electronics, Vol. 52, Part 1 (November 1953), 593-99.

- The map is a diagram made up of squares. Each square represents one minterm.
- Since any Boolean function can be expressed as a sum of minterms, it follows that a Boolean function is recognized graphically in the map from the area enclosed by those squares whose minterms are included in the function.
- We shall assume that the simplest algebraic expression is any one in a sum of products or product of sums that has a minimum number of literals.
(This expression is not necessarily unique.)

3.2 Two- and Three-variable Maps

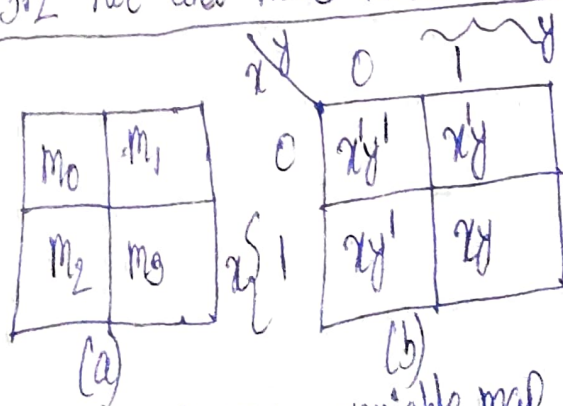


Figure 3.1 Two-variable map